Menelaus Theorem (without proof)

Statement

A necessary and sufficient condition for points P, Q, R on the respective sides BC, CA, AB (or their extension) of a triangle ABC to be collinear is that $\frac{BP}{PC} \times \frac{CQ}{QA} \times \frac{AR}{RB} = -1$ where all segments in the formula are directed segments.

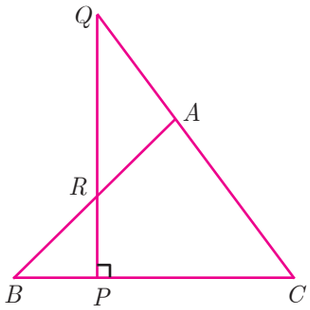


Fig. 4.71

Menelaus

Menelaus was a Greek mathematician who lived during the Roman empire in both Alexandria and Rome during first century (CE). His work was largely on the geometry of spheres.

Menelaus theorem was first discussed in his book, sphaerica and later mentioned by Ptolemy in his work Almagest.

Menelaus theorem proves that spheres are made up of spherical triangles.

Note

- Menelaus theorem can also be given as $BP \times CQ \times AR = -PC \times QA \times RB$.
- If BP is replaced by PB (or) CQ by QC (or) AR by RA, or if any one of the six directed line segments BP, PC, CQ, QA, AR, RB is interchanged, then the product will be 1.

Example 4.32 Show that in a triangle, the medians are concurrent.

Solution Medians are line segments joining each vertex to the midpoint of the corresponding opposite sides.

Thus medians are the cevians where D, E, F are midpoints of BC, CA and AB respectively.

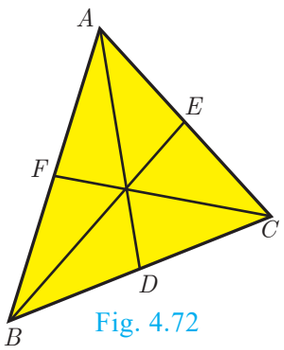


Fig. 4.72

Since D is a midpoint of BC, $BD = DC$ so $\frac{BD}{DC} = 1$ ...(1)

Since, E is a midpoint of CA, $CE = EA$ so $\frac{CE}{EA} = 1$ ... (2)

Since, F is a midpoint of AB, $AF = FB$ so $\frac{AF}{FB} = 1$ ... (3)

Thus, multiplying (1), (2) and (3) we get,

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1 \times 1 \times 1 = 1$$

And so, Ceva's theorem is satisfied.

Hence the Medians are concurrent.

Do You Know?

Centroid is the point of concurrence of the median of a triangle.

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